

# Text as Pieces

Project: Transparent Book Recommender

## 1. Recall and choose your starting point

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

### Feedback Loops

Without opening the old lesson, explain one key idea from Feedback Loops. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Nearest Neighbor Reasoning

Without opening the old lesson, explain one key idea from Nearest Neighbor Reasoning. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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### Sampling and Representation

Without opening the old lesson, explain one key idea from Sampling and Representation. Give an example and a mistake that the idea helps you avoid.

**My remembered idea, example and correction**

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## 2. Practise with fading help

Use W2: Encode and decode for Tokenisation and vocabulary. Record the answer once; the lesson and workbook refer to the same attempt.

**Given inputs · Procedure I chose · Middle steps · Result and check**

**Given inputs:** \_\_\_\_\_

**Procedure I chose:** \_\_\_\_\_

**Middle steps:** \_\_\_\_\_

**Result and check:** \_\_\_\_\_

\_\_\_\_\_

**The step I completed without help, and the hint I needed**

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## 3. Individual reasoning check

Use the worked case for Tokenisation and vocabulary. Proposed explanation: “A token always equals a word”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

**My explanation before partner discussion**

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## 4. Independent transfer case

## Independent case: Trace a new input independently

Under the original vocabulary encode sat cats, decode it and count text tokens. Then trace sat cat! and state the exact failure position using zero-based character positions.

### My independent explanation, trace and limits

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Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

## 5. Your learning target

Toy tokenizer A splits 'unhelpful' into un/help/ful; B keeps it whole. Count tokens for each and explain why a three-word sentence need not use three tokens.

## 6. Solve the given case

Toy tokenizer A splits rainbows as [rain][bow][s]; B uses [rainbows].

Count tokens and state the limit of the example.

Show your trace, calculation or evidence

### Write or draw your reasoning and evidence.

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## 7. Check a changed case

A classroom tokenizer splits sunflower into sun and flower. How many pieces result? Must every real tokenizer split it the same way?

My independent answer and reason

**Write or draw your reasoning and evidence.**

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### Week 25 Test and reflect

## 8. Design a revealing test

Create a changed input, boundary case or missing-information case. Write the expected answer or safe response before testing.

### Test record · My evidence

**Test record:** Changed input or condition

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Expected answer or fallback

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Observed result

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

**Test record:** Why the result occurred

**My evidence:** \_\_\_\_\_  
\_\_\_\_\_

## 9. Project checkpoint

Tokenize book descriptions before any text-based matching, while preserving the original source text.

The project change and evidence I will keep

**Write or draw your reasoning and evidence.**

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## 10. Self check

I can explain the mechanism. I can show evidence. I can handle a changed case. I can name a limit. Mark each as not yet, with help or independently, and attach supporting work.

Teacher review \_\_\_\_\_ Next step \_\_\_\_\_

## 11. Transfer practice from the original programme

Toy tokenizer A splits 'unhelpful' into un/help/ful; B keeps it whole. Count tokens for each and explain why a three-word sentence need not use three tokens.

**My independently reasoned answer**

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Record support used: none, hint or teacher explanation.

## 12. W1 Read the contract

Explain why sat wins over s at the start of sat and why uppercase C is unsupported.

**My reasoning and evidence**

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## 13. W2 Encode and decode

Encode cats sat using the given vocabulary, decode it, and count with and without an appended end marker.

**My reasoning and evidence**

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## 14. W3 Compare sequence lengths

Encode cats sat with character tokens only. Explain the difference from four text tokens in the original vocabulary.

**My reasoning and evidence**

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## 15. W4 Trace a new input independently

Under the original vocabulary encode sat cats, decode it and count text tokens. Then trace sat cat! and state the exact failure position using zero-based character positions.

### My reasoning and evidence

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## 16. W5 Audit silent repair

An encoder lowercases Cat and removes punctuation before encoding, but labels its output an exact encoding of the original. Explain what must be recorded or changed.

### My reasoning and evidence

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## 17. Feedback, return check and learning record

Evidence · My record

**Evidence:** Worked alone / hint / model

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** First step I repaired and why

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** New case checked later; date and result

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** What I can now explain without notes

**My record:** \_\_\_\_\_

\_\_\_\_\_

**Evidence:** One remaining question

**My record:** \_\_\_\_\_

\_\_\_\_\_

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

## 18. Pictorial case practice: Text as Pieces

A fictional tokenizer converts a short catalogue phrase into tokens.

Vocabulary entries: book, books and the space character. At each position choose the longest matching entry; preserve the actual space.

Text is “books book”. Exclamation marks have no token; zero-based character positions are used.

## Text as Pieces

### Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



#### 1. Read prefix

At position 0, longest matching entry is books, not book.



#### 2. Encode phrase

Tokens: books, space-character, book; three tokens.



#### 3. Decode

Joining token text returns exactly books book.



#### 4. Unknown character

For books book!, the ! at position 10 has no token: stop with error.

### CHANGE ONE THING

Use a character vocabulary covering b,o,k,s and space instead.

Explain the changed result using the matching lesson.

Why is the unknown ! at position 10 rather than position 9?

### My picture-based explanation and changed-case prediction

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Support used: alone / hint / worked example. This record is practice evidence.

Project connection: Document the recommender's text representation and unknown-token behavior rather than assuming every model counts words identically.